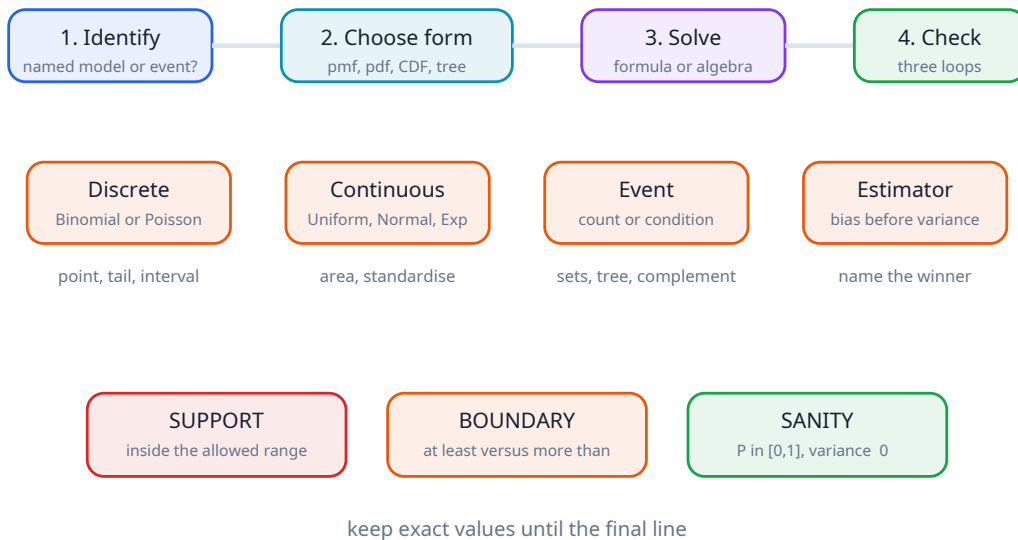


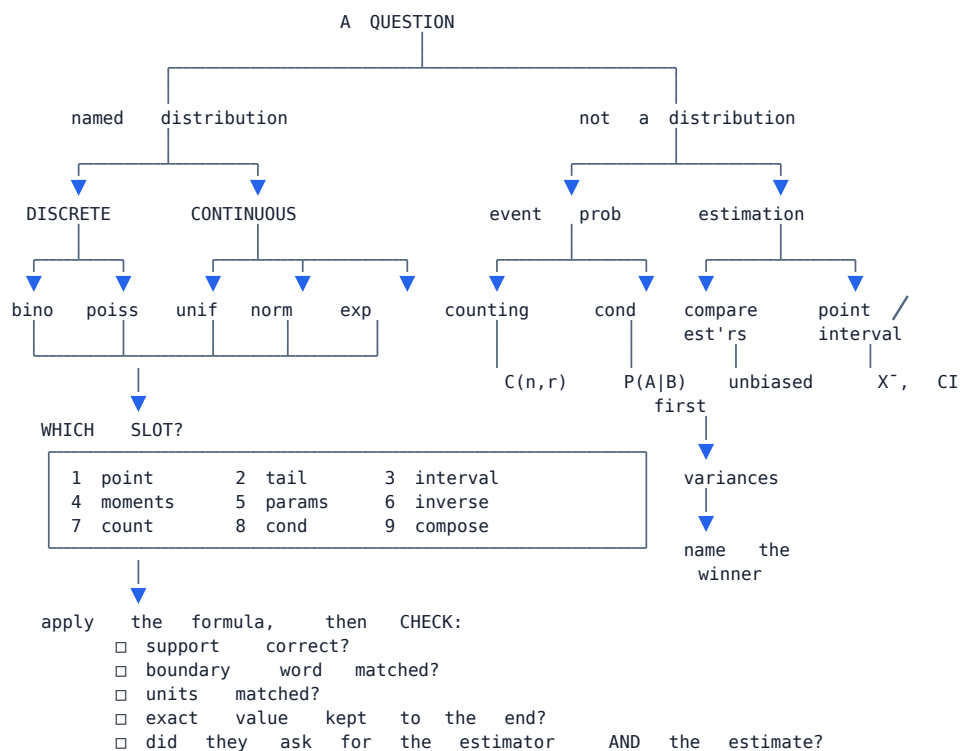
04 METHODS: solving path per question type

THE MASTER DECISION TREE

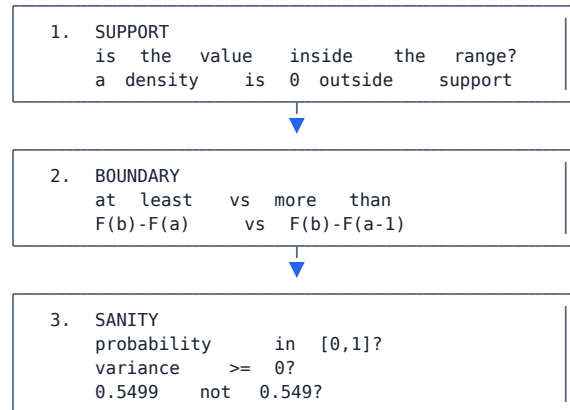
Question-solving decision path



Question-solving decision path



THE THREE CHECK LOOPS AFTER ANY ANSWER



Every method here is the one the course teaches, plus the fastest valid alternative. Each entry: when to use it, the steps, and the alternate with its tradeoff.

1. Probability of an event

when	any "find the probability that ..." that is not a named distribution
steps	1 name the sample space and count its size (ordered if the count needs it)
2 write the event as a set of outcomes, or as a complement	
3 if complement is easier, use $P(A) = 1 - P(A')$	
4 for "at least one" always go through the complement	
alternate	direct counting vs complement. Complement wins whenever the event says
"at least", "none", "all of them". Direct counting wins for small explicit sets.	

2. Conditional and independence

when	"given that", "if ... what is the probability", or a two-stage draw
steps	1 identify the condition B
2 write $P(A B) = P(A \cap B)/P(B)$	
3 get the intersection from the multiplication rule if the story gives conditionals	
alternate	a tree diagram. Slower to draw but it stops the $P(A B)$ vs $P(B A)$ swap.
independence test	$P(A \cap B) = P(A)P(B)$, or $P(A B) = P(A)$. Either form, one line.

3. Build a distribution (pmf or pdf) from a story

when	"find the probability distribution of X", or "X has pdf $f(x)=\dots$ "
steps	1 write the SUPPORT of X first
2 for discrete, tabulate x and $p(x)$, then verify $\text{sum} = 1$	
3 for continuous, write $f(x)$ with its range, then verify $\text{integral} = 1$	
alternate	if the story is a named distribution, skip the build and name it (binomial,
Poisson, etc), then quote pmf and moments.	

4. Find the constant k

when	"find k such that $f(x)$ is a density" or "determine k"
steps	1 write the normalisation, integral (or sum) over the SUPPORT = 1
2 integrate the polynomial, collect k	
3 solve for k	
traps	the support is finite here, not $(-\infty, \infty)$. A scheme that prints infinite bounds and
then evaluates on a finite support is printing a slip (see errata 20).	
the sum version: watch for a k^2 term, the equation can be quadratic with one valid	
positive root (the batch-1 assignment gives $k = 1/10$ from $10k^2 + 9k = 1$)	

5. cdf from pmf or pdf, and back

when	"find the cdf", "find $F(x)$ ", "given $F(x)$ find $f(x)$ ", "find $P(X < 6)$ from the cdf"	
steps	pmf $\rightarrow$ cdf	accumulate: $F(x) = \text{sum of } p \text{ up to } x$. Piecewise on the INTEGER values.
pdf $\rightarrow$ cdf	integrate f from the lower support edge to x	
cdf $\rightarrow$ pdf	differentiate F on each interval	
cdf $\rightarrow$ pmf	the jump $F(x) - F(x-)$	
traps	a cdf accumulates ALL earlier mass. For a piecewise cdf on $1 \leq x \leq 2$ you must add	
the mass from 0 to 1 (errata 19: the 2024 MTE CDF is option D, not the scheme's B)		
the minimum c with $F(c) > 1/2$ is a cdf query, read it off the accumulation		

6. Expectation

when	"find $E(X)$ ", "find the expected value", "find $E[h(X)]$ "
steps	1 write the sum (discrete) or integral (continuous), over the support
2 plug, integrate or sum	
3 for $E(X^2)$ repeat with x^2	
alternate	symmetry: a density symmetric about c has $E(X) = c$ with no integration.
traps	$E[g(X)] \neq g(E(X))$. Compute the integral of g , do not plug E into g .
for a count of items: expected count = $N \times P$, do not report P alone.	

7. Variance

when	"find the variance", "find the sd", "find $E(X)$ and $\text{Var}(X)$ "
steps	1 $E(X)$
2 $E(X^2)$ from its own sum/integral	
3 $\text{Var} = E(X^2) - [E(X)]^2$	
4 sd = $\sqrt{\text{Var}}$ when asked for spread in original units	
alternate	for a named distribution quote np , npq , l , $(b-a)^2/12$, $1/l^2$ directly.
traps	keep EXACT values until the final line (errata 12: rounding Var then rooting it)
gives 0.975, the exact sd is 0.9734 $\rightarrow$ 0.97)	
the variance is never negative, if you get a negative check your arithmetic	

8. Chebyshev

when	"using Chebyshev's inequality", "find the lower bound", "without assuming a
distribution", "is this distribution-free"	
steps	1 get μ and σ (compute if not given, $\sigma = \sqrt{\text{Var}}$)
2 convert the event to $ X - \mu \geq \epsilon$ or $< \epsilon$	
3 $k = \epsilon / \sigma$	(raw units to sigma units)
4 tail form $1/k^2$, or complement form $1 - 1/k^2$, whichever the wording wants	
alternate	if the exact distribution is known, compute the exact value TOO and report
"bound 35/54, exact 1/3, the bound holds and is wider"	
traps	k must be in sigma units. The complement form is for "at least"/"within".
The bound can exceed 1 ($k \leq 1$), then it is vacuous.	
errata 15: the L10-11 deck's Q3 slide mis-states its own row, use Q1 and Q2 only.	

9. Binomial

when	n trials, two outcomes, independent, p constant, count the successes
steps	1 state $X \sim B(n, p)$, write $q = 1-p$
2 the ask: point, tail (complement), or interval	
3 substitute the pmf, use $C(n, x)$	
alternate	for "at least 1" use $1 - q^n$ (faster than summing)
traps	"at least 2" = $1 - P(0) - P(1)$, the classic two-term complement
read "p is the success probability", if the story gives failures, invert it first	

10. Poisson

when	rare events, arrivals, in a fixed window, or "use the approximation" with n big p small
steps	1 get l. If given a rate, scale it to the window. If given np in an approximation,
$l = np$.	
2 the ask: point $e^{-l} l^x / x!$, tail (complement), or the zero case e^{-l}	
3 state $\text{Var} = l$ if asked, it equals the mean	
alternate	for the ratio question $P(X=1) = c P(X=2)$, the e^{-l} cancels, leaving $l = 2/c$
traps	rate vs window scaling (2 per minute, then 5 minutes $\rightarrow l =$ the window's rate)
the nesting: Poisson per minute THEN binomial over the minutes (0.00145 question)	
a Poisson variable has equal mean and variance, but equality alone does not prove	
an otherwise unspecified distribution is Poisson	

11. Uniform

when	"equally likely in an interval", "random point"
steps	1 identify a and b
2 probability = length ratio, $P(a \leq X \leq b) = (\text{interval}) / (b-a)$	
3 moments: mean $(a+b)/2$, variance $(b-a)^2/12$	
traps	clip to the support, a value outside $[a, b]$ has probability 0 or 1, not the ratio
absolute-value events become intervals first: $ X-5 < 1$ is $4 < X < 6$	

12. Normal

when	normal, approximately normal, large-sample mean, or the CLT destination
steps	1 identify μ and σ (σ from variance, sqrt it)
2 DECIDE which table the question gives (ϕ area-from-0, or F cumulative)	
3 standardise $Z = (X - \mu) / \sigma$	
4 read the table, apply symmetry $\Phi(-z) = 1 - \Phi(z)$	
5 for the inverse: get z for the probability, then $X = \mu + z \sigma$	
alternate	landmarks: within 1s 68.27, 2s 95.45, 3s 99.73, for a fast sanity check
traps	the second parameter is VARIANCE, sqrt it
the inverse's forgotten last step: convert z back to the X scale	
two-unknown questions: two given probabilities give two equations, solve simultaneously	
(errata 10: use $\mu = 37.2$, $\sigma = 28.2$, cutoff = 30.4)	

13. Exponential

when	waiting time, time between arrivals, lifetime, "no memory"
steps	1 get λ (a RATE). If given a mean, $\lambda = 1/\text{mean}$.
2 match the units of λ and t	
3 point/interval: $F = 1 - e^{-\lambda t}$.	tail: $e^{-\lambda t}$.
4 memoryless if the condition is on surviving	
alternate	mean-first form $f = (1/\mu) e^{-t/\mu}$ when the question gives μ directly
traps	rate vs mean reciprocal (errata 4: the slide p040 states it backwards, read p041)
conditional in the exponential: $P(X < 1 \mid X < 2) = (1 - e^{-0.5}) / (1 - e^{-1}) = 0.6225$	
(errata 9: the assignment prints 0.5679, that is wrong)	

14. Sampling and standard error

when	"standard error", "sampling distribution", "the mean of a sample of n "
steps	1 $SE = \sigma / \sqrt{n}$ (or $S / \sqrt{n}$ if the sample sd is given)
2 the "average of n " template divides by SE, not σ	
alternate	the sqrt- n rule for sample-size questions: quadruple n halves SE
traps	a single observation uses σ ; an average of n uses $\sigma / \sqrt{n}$. Mixing these
is the commonest CLT slip.	

15. CLT

when	"the sample mean is approximately normal", n listed, population shape unknown
steps	1 if the population is normal, $\bar{X}$ is exactly normal; otherwise assess whether n and
the population shape make the CLT approximation reasonable	
2 $\bar{X} \sim N(\mu, \sigma^2/n)$, so $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$	
3 read the table as a normal	
alternate	for a SUM not a mean, the sum is $N(n\mu, n\sigma^2)$, σ scales by $\sqrt{n}$ not n
traps	$n < 30$ with a non-normal population: do not use the normal approximation automatically
$n \geq 30$ is the course heuristic, not a theorem boundary	
a normal population is exact at any n , this is the fallback for small- n questions	
(errata 6: impurity $Z_2 = -0.94$, the slide prints -0.4)	

16. Estimator comparison (the 6 to 8 mark question)

when	"which estimator is unbiased", "most efficient", "find the constant"
steps	1 write E of each estimator, expand using linearity
2 unbiased: $E(T) = \theta$. Solve for the constant if one exists.	
3 for each UNBIASED estimator, compute $\text{Var}(T)$ using $\text{Var}(X_i) = \sigma^2$ and independence	
4 the smallest variance wins	
alternate	relative efficiency $\text{Var}(T_2)/\text{Var}(T_1)$ when asked to compare a pair
traps	a biased estimator is DISQUALIFIED before the variance comparison, even if its
variance is smaller (errata 18, mock B4: T_4 is biased and third of four, the note	
calling it "smallest" is wrong)	
consistency is a large- n property, a fixed- n small variance does not make a	
consistent estimator	

17. The protocol when a question chains parts

when	"find k ... hence find the mean ... hence find $P(X > \dots)$ "
steps	1 treat each part as independent given the previous answer
2 carry the EXACT value forward, round only at the final line	
3 if a part is impossible, say so, do not silently continue (the asgn-variant has	
a row and a key that cannot both be right, errata 5.1)	